Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

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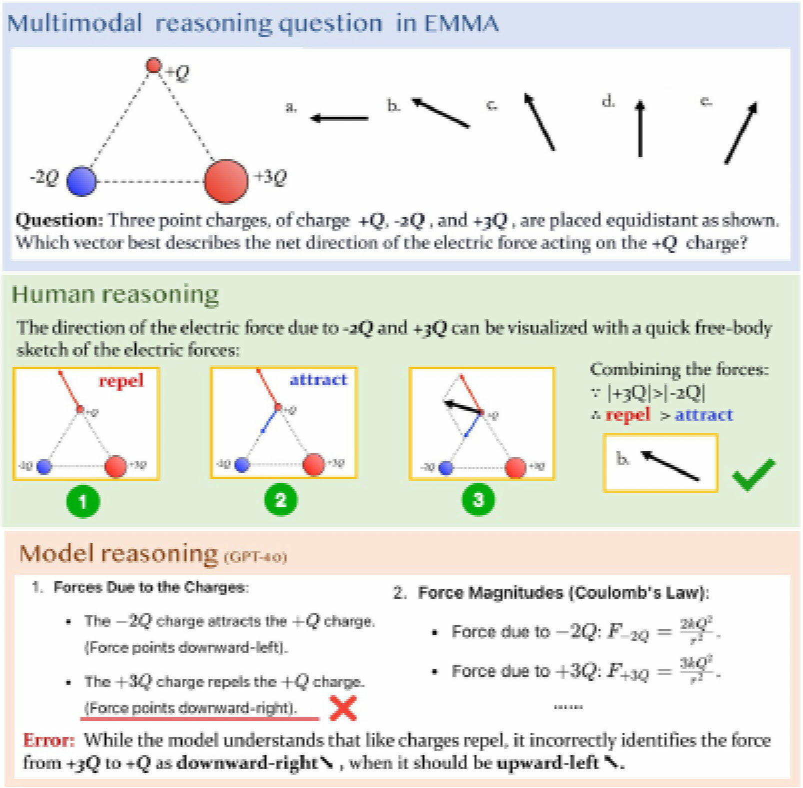
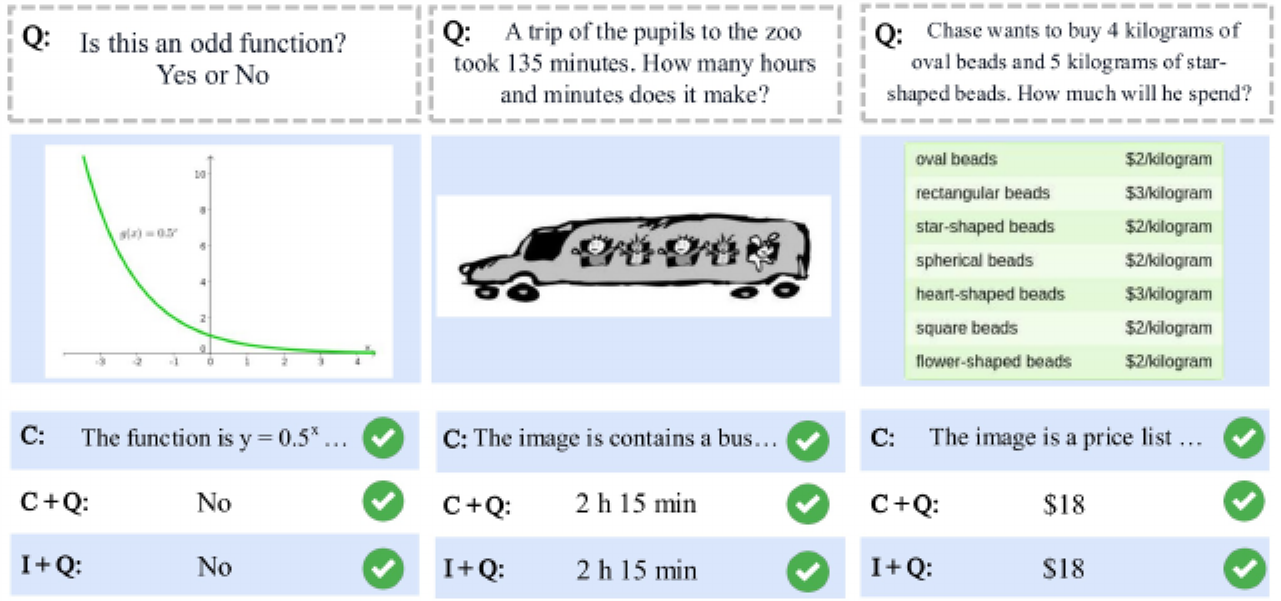
01 Motivation And Background

- Human reasoning integrates language and vision for spatial, physical, and diagrammatic tasks
- LLMs excel in text; MLLMs' true cross-modal reasoning remains unclear
- Existing benchmarks emphasize recall or shallow perception over integrated reasoning
- Gap: evaluating iterative visual–text interplay and multi-step multimodal reasoning



01 Limitations Of Current Benchmarks

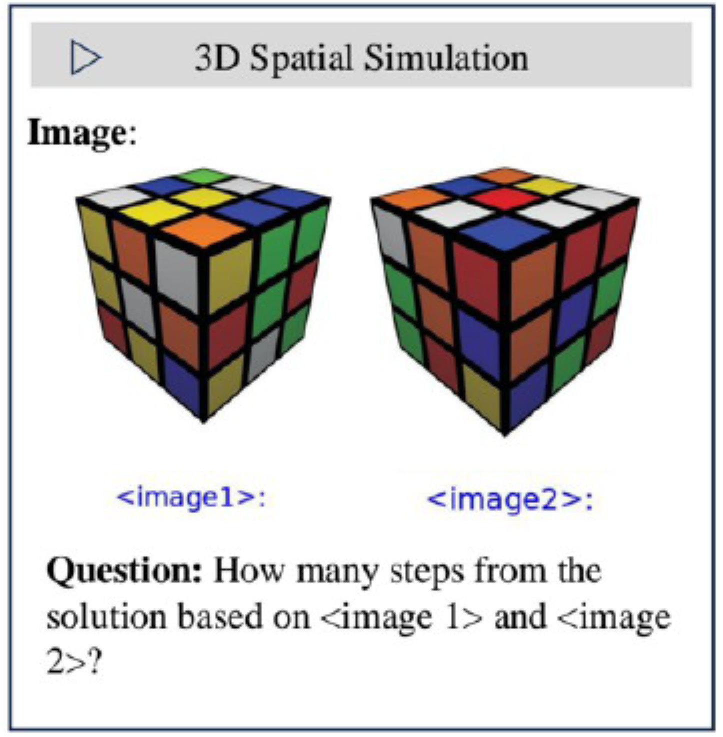


- Datasets allow language-only shortcuts or single visual pass solutions
- Filtering removes text-only items but leaves perception-aided, not reasoning-required problems
- Evaluations conflate perception with deeper visual simulation and integration

02 MLLM Progress And Visual CoT

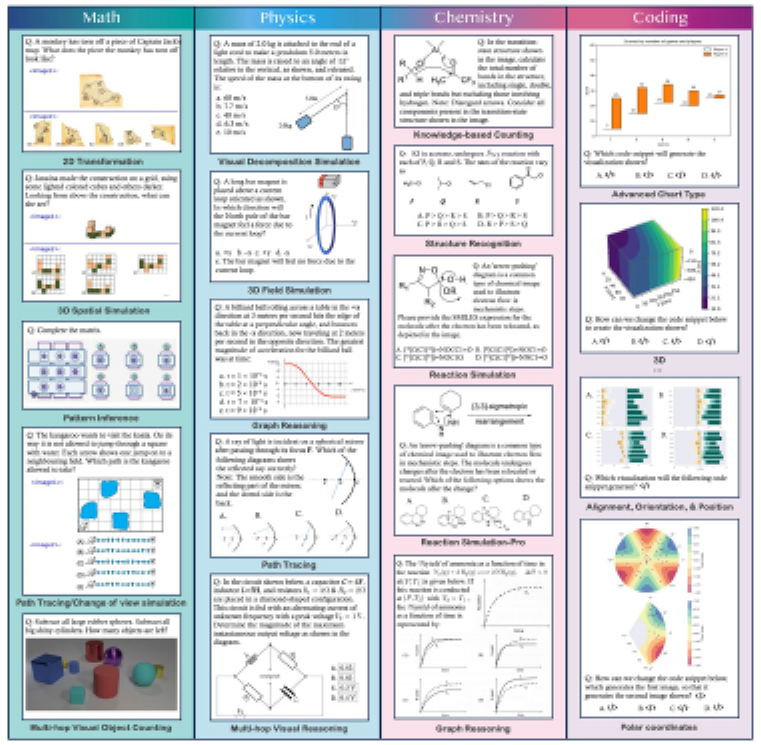
- MLLMs gain from visual instruction tuning, showing strong perception and some reasoning
- Visual CoT improves attention to regions/attributes but not visual manipulation or simulation
- Complex STEM tasks require cross-modal multi-hop reasoning beyond current methods

02 Benchmarks And Their Shortcomings



- Redundancy between text and images enables bypassing visual reasoning
- MMMU-Pro improves filtering, but caption-plus-text can still solve items
- Need a benchmark requiring intrinsic visual reasoning and iterative cross-modal steps

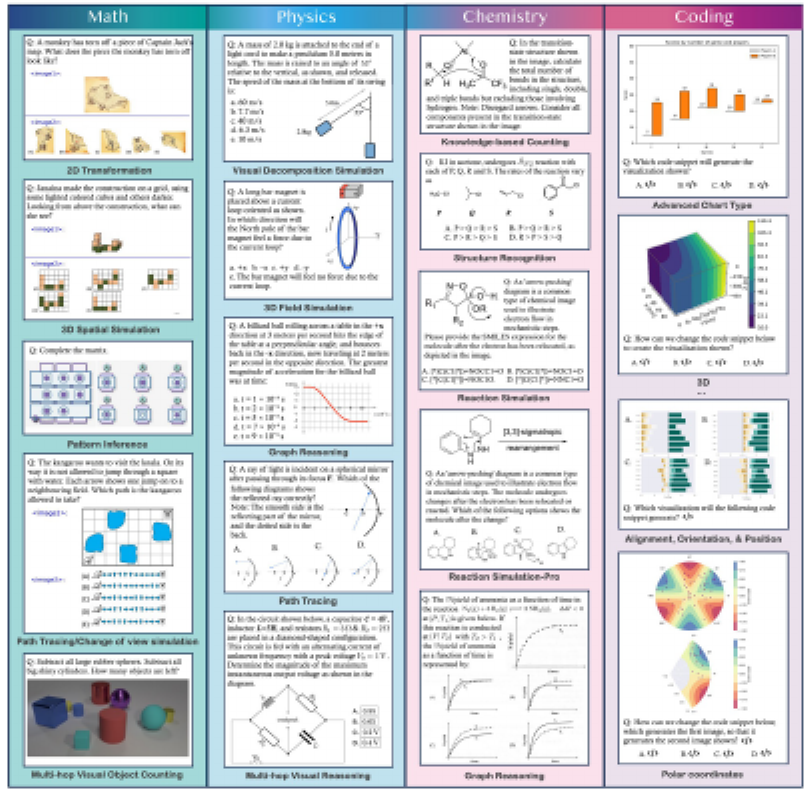
03 An Enhanced Multimodal Reasoning Benchmark



- EMMA: 2,788 problems across math, physics, chemistry, and coding
- Tasks unsolvable by text-only or single visual pass; require iterative cross-modal reasoning
- Concise answers enable reliable automated evaluation

03 Fine-Grained Skill Taxonomies And Labels

- Problems labeled for specific multimodal skills enabling granular diagnostics
- Covers patterns, transformations, decomposition, simulations, spatial visualization
- Expert-created/verified labels guide targeted capability research

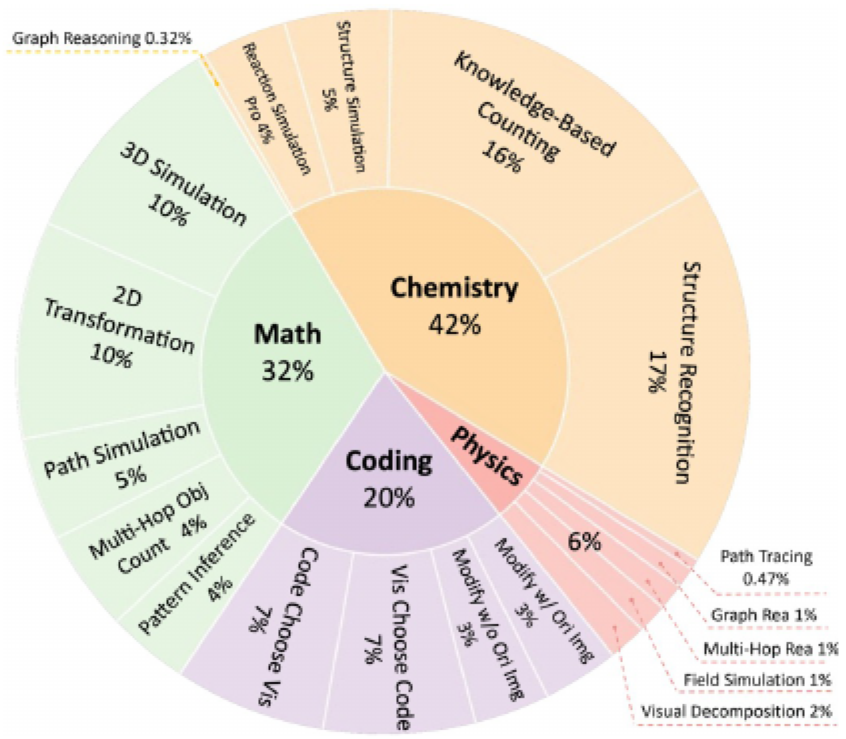


03 Stricter Filtering And Substantial New Data

Statistic	Number
Total questions	2,788
- Multiple-choice questions	2,002 (72%)
- Free-form questions	786 (28%)
- Questions with answers	2,788 (100%)
- Questions newly added	1,796 (64%)
Image in the question	2,599 (93%)
Image in the option(s)	195 (7%)
Problems with multiple images	298 (10%)

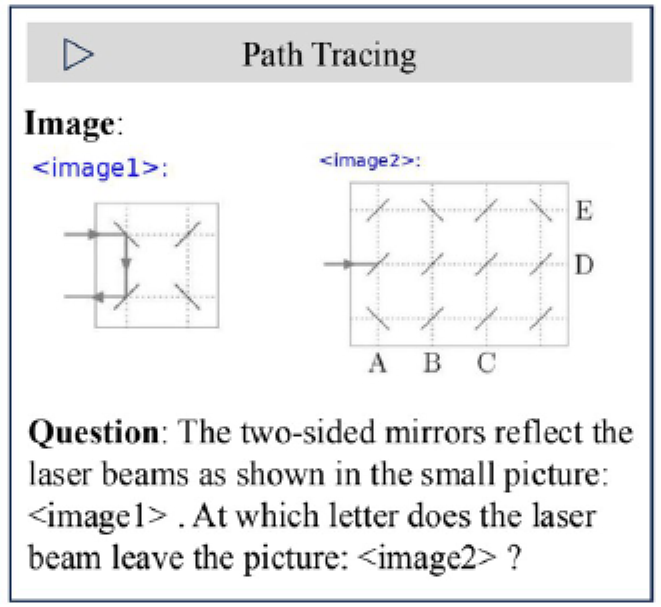
- Caption-augmented filter removes caption-plus-text solvable items
- Adds 1,796 new problems, expanding physics, chemistry, and coding
- Introduces standardized visualization-coding MCQ evaluation

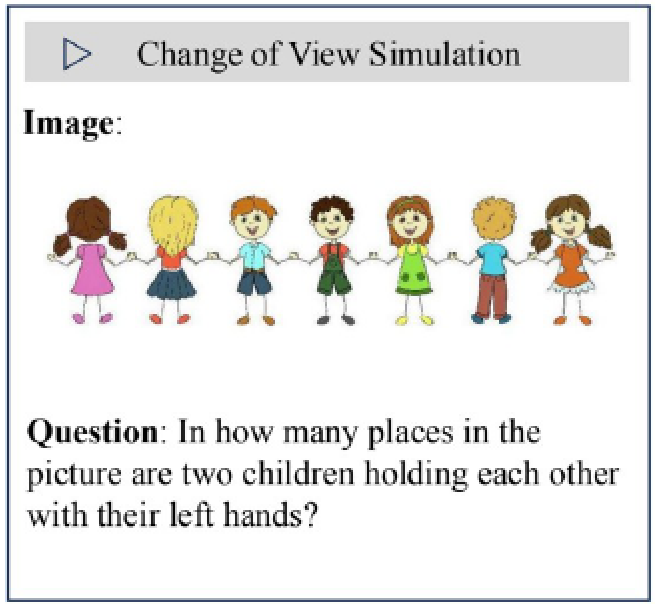
04 Dataset Composition And Formats



- 2,788 questions: 2,002 multiple-choice, 786 free-form
- Images in 93% of questions; 10% have multiple images
- EMMA-mini: 400 balanced questions for efficient comparisons

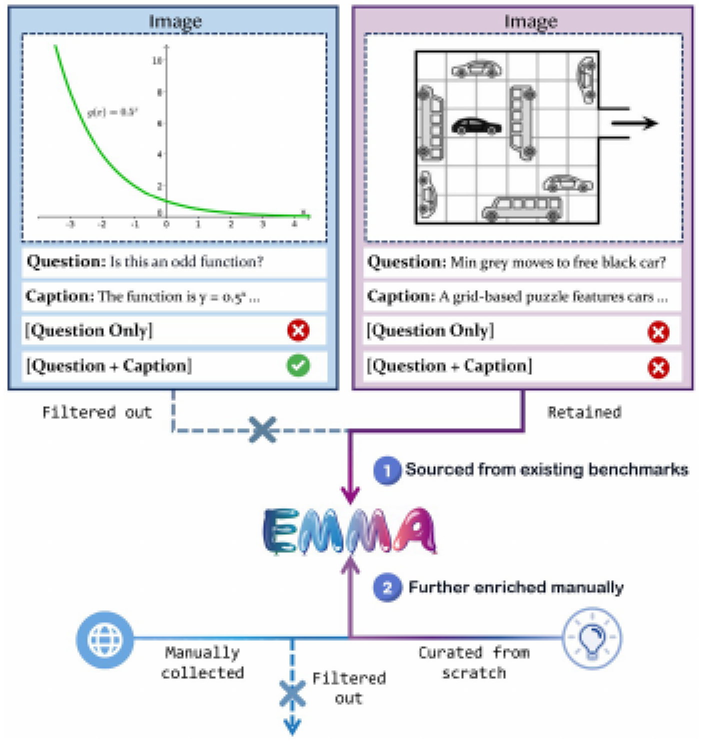
04 Skill Coverage Across Subjects





- Math: 3D simulation, 2D transformation, path tracing, counting, pattern inference
- Physics: 3D field simulation, graph reasoning, path tracing
- Chemistry: structure recognition, bond counting, reaction simulation

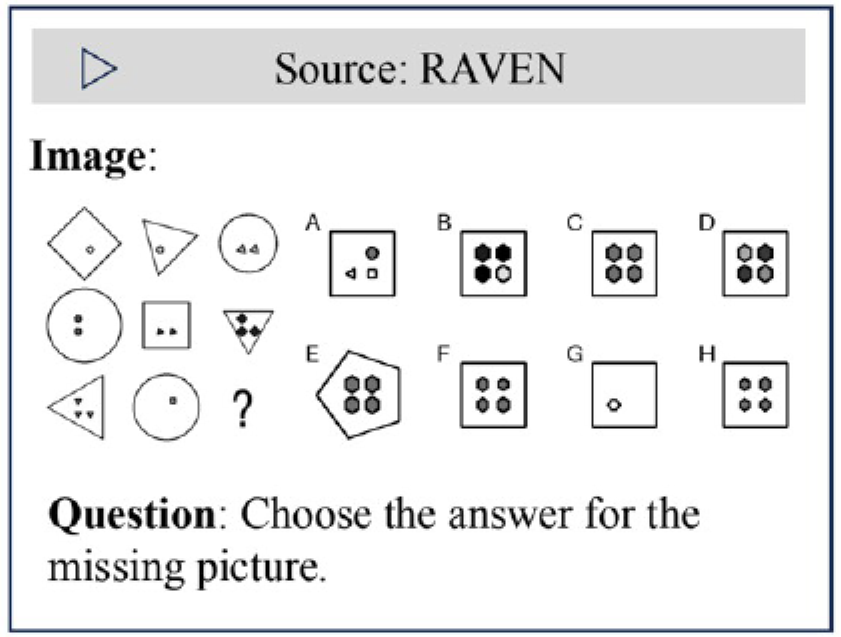
05 Step 1: Source And Caption-Augmented Filtering



- Source from prior benchmarks and web materials
- Caption images; provide text plus captions to models
- Discard items any model solves ≥5/10 across 30 trials
- Removes text-only and single-pass solvable problems

05 Step 2: Taxonomy Labeling And Expansion

- Categorize remaining items into fine-grained skill taxonomies
- Manually construct problems to cover underrepresented skills
- Chemistry via RDKit; reactions with expert annotations
- Coding seeded from curated visualizations with controlled variations



05 Subject-Specific Construction Details

- Math: refine MathVision/MathVista; add RAVEN for multi-hop reasoning
- Physics: aggregate OlympiadBench, EXAMS-V, MMMU, curated AP/Khan
- Chemistry: emphasize structures and reactions beyond formulas
- Coding: four visualization tasks via CharXiv and matplotlib exemplars

06 Data Splits And Human Baseline

- EMMA-mini: 100 balanced questions per subject for fast iteration
- Two experts per subject evaluate; average provides human baseline
- Benchmarks model gaps in multimodal reasoning

06 Models And Prompting Strategies

Model	Parameter Setting	Source	URL
GPT-4o pass@1	temperature = 0.0	chatgpt-4o-latest	https://platform.openai.com
GPT-4o pass@n	temperature = 0.7	chatgpt-4o-latest	https://platform.openai.com
Claude 3.5 Sonnet	temperature = 0.0	claude-3-5-sonnet	https://www.anthropic.com/
Gemini 2.0 Flash pass@1	temperature = 0.0	gemini-2.0-flash-exp	https://ai.google.dev/
Gemini 2.0 Flash pass@n	temperature = 0.7	gemini-2.0-flash-exp	https://ai.google.dev/
Gemini 2.0 Flash Thinking-1219	temperature = 0.0	gemini-2.0-flash-thinking-exp-1219	https://ai.google.dev/
Gemini 2.0 Flash Thinking-0121	temperature = 0.0	gemini-2.0-flash-thinking-exp-0121	https://ai.google.dev/
OpenAI o1	-	interface	https://chatgpt.com/
Qwen2-VL-72B-Instruct	temperature = 0.7	local checkpoint	https://huggingface.co/Qwen/Qwen2-VL-72B-Instruct
QVQ-72B-Preview	temperature = 0.7	local checkpoint	https://huggingface.co/Qwen/QVQ-72B-Preview
LLaVA-Onevision-72B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/llava-hf/llava-onevision-qwen2-72b-ov-hf
InternVL2-Llama3-76B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/OpenGVLab/InternVL2-Llama3-76B
InternVL2.5-78B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/OpenGVLab/InternVL2_5-78B

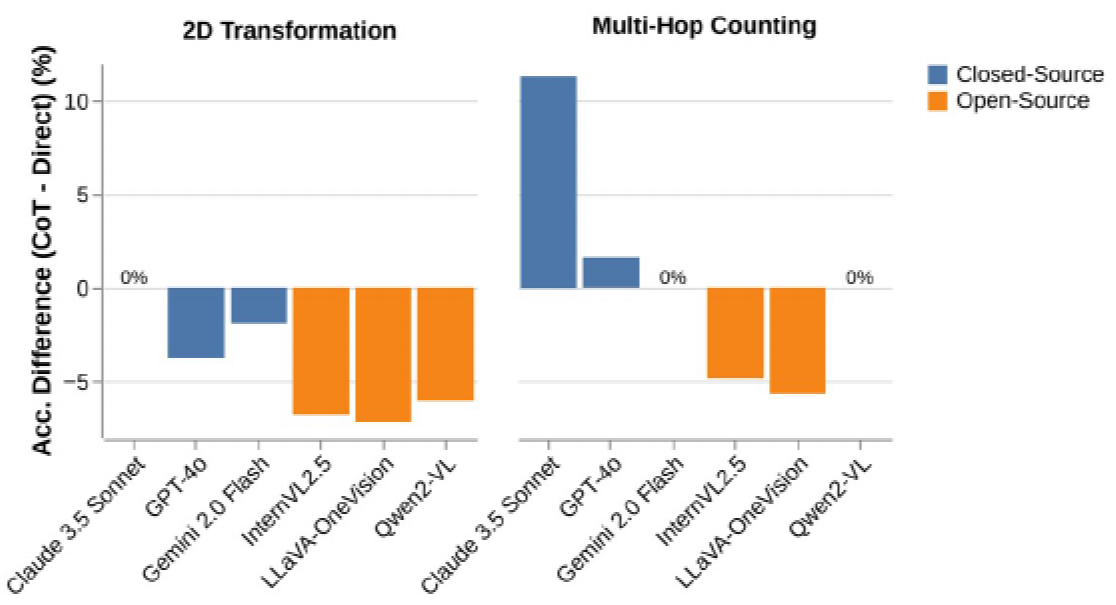
- Evaluate ten SoTA MLLMs: open-source and proprietary
- Two prompting modes: direct and Chain-of-Thought
- Assess impact of explicit textual reasoning

07 Overall Performance And Gaps To Humans

	CoT	EMMA					EMMA-mini				
		Math (892)	Phys. (156)	Chem. (1,176)	Coding (564)	Overall (2,788)	Math (100)	Phys. (100)	Chem. (100)	Coding (100)	Overall (400)
Random choice	–	14.01	25.64	16.50	25.71	18.08	13.00	23.00	27.00	28.00	22.75
Human Expert	–	–	–	–	–	–	75.00	64.50	86.00	85.50	77.75
Claude 3.5 Sonnet	✗	25.34	33.97	40.90	38.65	35.08	23.00	34.00	44.00	35.00	34.00
Gemini 2.0 Flash	✗	23.88	38.46	36.31	42.02	33.61	20.00	40.00	36.00	41.00	34.25
GPT-4o	✗	27.24	38.46	31.89	40.07	32.42	30.00	38.00	33.00	40.00	35.25
Qwen2-VL-72B-Instruct	✗	33.07	42.31	32.06	34.57	33.46	38.00	40.00	34.00	37.00	37.25
LLaVA-Onevision-72B	✗	27.69	35.90	25.26	28.72	27.33	25.00	32.00	24.00	28.00	27.25
InternVL2-Llama3-76B	✗	25.11	22.44	24.06	27.84	25.07	31.00	22.00	21.00	28.00	25.50
InternVL2.5-78B	✗	31.39	38.46	35.20	31.91	33.50	30.00	40.00	38.00	33.00	35.25
Claude 3.5 Sonnet	✓	29.37	41.03	41.07	40.60	37.23 (↑ 2.15)	30.00	38.00	41.00	39.00	37.00 (↑ 3.00)
Gemini 2.0 Flash	✓	25.90	38.46	24.66	40.96	29.12 (↓ 4.48)	24.00	41.00	36.00	44.00	36.25 (↑ 2.00)
GPT-4o	✓	25.56	43.59	33.67	39.01	32.71 (↑ 0.29)	27.00	44.00	35.00	38.00	36.00 (↑ 0.75)
Qwen2-VL-72B-Instruct	✓	27.69	34.62	24.57	29.43	27.12 (↓ 6.35)	35.00	34.00	32.00	23.00	31.00 (↓ 6.25)
LLaVA-Onevision-72B	✓	22.42	15.38	22.70	30.67	23.82 (↓ 3.52)	23.00	26.00	23.00	29.00	25.25 (↓ 2.00)
InternVL2-Llama3-76B	✓	22.20	32.05	19.73	30.32	23.35 (↓ 1.72)	27.00	33.00	21.00	32.00	28.25 (↑ 2.75)
InternVL2.5-78B	✓	25.56	39.74	27.47	25.18	27.08 (↓ 6.42)	31.00	36.00	24.00	19.00	27.50 (↓ 7.75)
Gemini 2.0 Flash Thinking-1219	–	31.61	56.41	37.93	43.44	38.06	35.00	57.00	41.00	41.00	43.50
Gemini 2.0 Flash Thinking-0121	–	**37.11**	**60.26**	**41.58**	**48.05**	**42.50**	34.00	**63.00**	**47.00**	48.00	**48.00**
QVQ-72B-Preview	–	–	–	–	–	–	34.00	39.00	24.00	31.00	32.00
o1	–	–	–	–	–	–	**41.00**	49.00	40.00	**53.00**	45.75

- All models underperform humans; best achieves 48.0% on EMMA-mini
- Human gap averages 29.75%; smallest in physics; large in math, chemistry, coding
- Closed-source generally lead; Gemini 2.0 Flash Thinking strongest

07 Effect Of Chain-Of-Thought Prompting



- Textual CoT helps many closed-source models
- Open-source models often degrade; >6% drops from direct prompting leaders
- CoT may induce hallucinations for tasks needing fine-grained visual simulation

08 Method 1: Majority Voting

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
	Majority Voting	–		–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
GPT-4o	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
	Majority Voting	–		–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
Gemini 2.0 Flash	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
	Majority Voting	–		–	48.00	49.00	**50.75**
Gemini 2.0 Flash Thinking	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

- Sample multiple CoT candidates; select most frequent answer
- Improves accuracy up to N=16 for stronger bases, still below humans
- Limited when visual reasoning steps are wrong or missing

08 Method 2: Best-Of-N With Reward Models

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
	Majority Voting	–		–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
GPT-4o	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
	Majority Voting	–		–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
Gemini 2.0 Flash	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
	Majority Voting	–		–	48.00	49.00	**50.75**
Gemini 2.0 Flash Thinking	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

- Select best among N using reward model; stronger external models work better
- Self-rewarding underperforms; improvements modest overall
- Reward models struggle with complex multimodal reasoning assessment

08 Method 3: Tournament Selection

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
	Majority Voting	–		–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
GPT-4o	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
	Majority Voting	–		–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
Gemini 2.0 Flash	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
	Majority Voting	–		–	48.00	49.00	**50.75**
Gemini 2.0 Flash Thinking	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

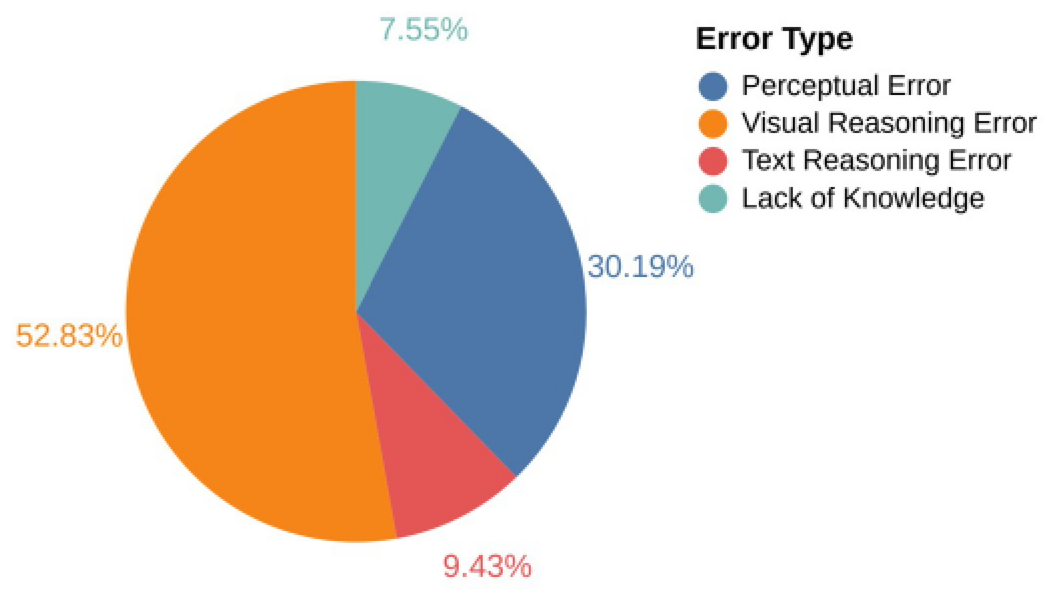
- Pairwise competitions adjudicated by strong reward model
- Outperforms Best-of-N via comparative judgments
- Still leaves sizable gap to human experts

08 Scaling Insights And Upper Bounds

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	–	36.00	–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
	BoN	Gemini Flash Thinking		40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	–	36.25	–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
	BoN	Gemini Flash Thinking		36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	–	43.50	–	48.00	49.00	**50.75**
	Tournament	Gemini Flash Thinking (Self)		45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

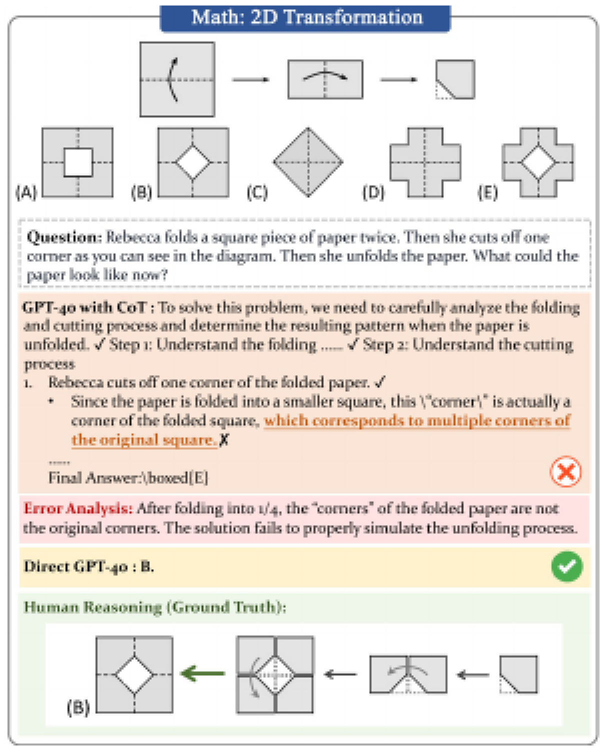
- Pass@N increases with stronger base reasoning; coverage improves
- Beyond N≈8, weaker models degrade
- Better base and reward models raise ceilings but cannot replace visual simulation

09 Visual Reasoning As The Bottleneck



- Errors concentrate in tasks needing precise spatial simulation and integration
- For o1 on EMMA-mini math/coding, visual reasoning errors exceed 50%
- Surpass perceptual, textual reasoning, and knowledge errors

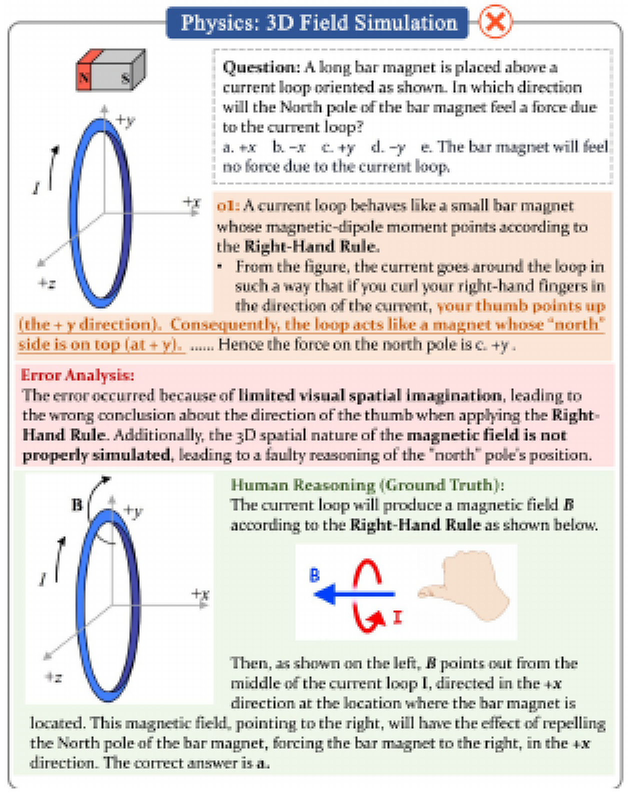
09 When Textual CoT Helps Or Hurts



- CoT hurts visually centric tasks like 2D transformations
- Helps tasks amenable to linguistic scaffolding, e.g., multi-hop counting
- Open-source models show increased hallucinations with CoT

09 Representative Failure Cases

- Models recall principles but mis-simulate spatial directionality
- Confusions in pattern/field tasks and skipped graphical decomposition
- Coding: miss subtle styling or scale differences



Conclusions And Takeaways

EMMA isolates genuine multimodal reasoning by eliminating shortcuts

Large human gaps; limited gains from CoT and scaling

Visual reasoning is the dominant bottleneck to address

Benchmark Limitations

Manual curation and balance may introduce biases; physics underrepresented

Future models may find new shortcuts; requires periodic refresh

Broaden chemistry topics and diversify sources for robustness

- Architectures enabling visual simulation, spatial manipulation, grounded visual CoT
- Richer supervision, diagram tool-use, integrated reward modeling
- EMMA sets a higher standard toward trustworthy multimodal reasoning

THANKS!